Set—**02**

Shahjalal University of Science and Technology
Department of Computer Science and Engineering
3rd year 2nd Semester Final Examination—June 2020 (Session 2017-18)
Course No.—**CSE 329**
Course Title—**Computer Architecture**
Time—**5 Hours** Credit: **3.00** Total Marks#**30**

(Answer All the Questions)

Group A

1. Answer the following **Five** Questions in short. $5 \times 1 = 5$
   (a) List the five classic components of a computer system.
   (b) What is the advantage of using virtual memory?
   (c) What is microarchitecture?
   (d) When underclocking is useful?
   (e) State Amdahl's Law

2. Answer the following **Two** Questions. $2 \times 2.5 = 5$
   (a) What do you mean by Control hazard? Briefly mention the common methods to overcome control hazards.
   (b) Why is a Single-Cycle Implementation not used today? What are the advantages of pipelined Datapath?

3. A compiler designer is trying to decide between two code sequences for a computer. The hardware designers have supplied the following facts:

	CPI for each instruction class		
	A	**B**	**C**
CPI	**1**	**2**	**3**

For a high-level language statement, the compiler writer is considering two code sequences that require the following instruction counts:

	Instruction counts for each instruction class		
Code Sequence	**A**	**B**	**C**
C-1	**2**	**1**	**2**
C-2	**4**	**1**	**1**

Which code sequence executes the most instructions? Which will be faster? What is the CPI for each sequence? 5

Group B

1. Answer the following **Five** Questions. $5 \times 1 = 5$
   (a) What do you mean by ISA?
   (b) What do you mean by opcode?
   (c) What do you mean by Instruction and Instruction set?
   (d) Draw the Memory hierarchy.
   (e) What is CPI?

2. Answer the following **Two** Questions. $2 \times 2.5 = 5$
   (a) Briefly describe the steps that transform a program written in a high-level language such as C into a representation that is directly executed by a computer processor.
   (b) Find out the decimal equivalent of the following floating-point binary number:

1	01111110	10000000000000000000000

3. What is the Data hazard? Explain with an example. Describe the following techniques with proper diagrams. 5
   - Pipeline Stall
   - Forwarding

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